

Software Requirements Specification

Version 1.0 | Confidential

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Document Type	SRS — Full Platform Specification
Target Market	India — Colleges, Universities, EdTech
Architecture	Azure Cloud Multi-Tenant SaaS
Total Modules	12 Modules across 3 Phases

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1. Document Overview

This Software Requirements Specification (SRS) defines the complete functional, technical, and business requirements for a cloud-based multi-college management SaaS platform targeting the Indian higher-education market. The platform is modeled on proven school-management systems and extends them with college-specific workflows, multi-tenancy, alumni networks, clubs management, and a campus marketplace.

Attribute	Detail
Project Name	Multi-College SaaS Platform
Version	1.0
Status	Draft
Platform Type	B2B SaaS — Cloud Hosted (Azure)
Target Market	Colleges, Universities, Polytechnics — India
Primary Language	English (UI); Hindi, Telugu, Tamil localisation in Phase 3

2. System Overview

The platform provides a centralised, browser-based and mobile-accessible environment for college administration, faculty, students, parents, and alumni. Each college is an independent tenant with isolated data, custom branding, and per-campus segregation. A super-admin layer allows the platform owner to manage all tenants, billing, and system health from a single dashboard.

Key Value Propositions

- Zero-installation, cloud-first — accessible from any browser or mobile device.
- Complete isolation — each college's data is private and never shared.
- Single mobile app for all roles: admin, faculty, student, parent, alumni.
- Integrated payments via Razorpay and PayU for Indian banking rails.
- WhatsApp and SMS alerts natively built in for Indian communication patterns.
- Clubs, alumni, and marketplace modules add long-term student engagement.
- Tier-based pricing from free (small colleges) to enterprise (universities).

3. Multi-Tenancy Architecture

Multi-tenancy is the foundational design principle. Every college is an independent tenant; no college can ever access another college's data.

Requirement	Implementation	Priority
Isolated database per tenant	Separate Azure SQL schema per college	MVP
Custom subdomain	college-name.platform.com via Azure Front Door	MVP
Branding per tenant	Logo, college name, colour theme stored per tenant	MVP
Super admin panel	Central dashboard: all colleges, billing, health	MVP
Per-tenant billing	Razorpay subscriptions, plan upgrades, invoices	MVP
Role isolation	JWT claims scoped to tenantId; enforced at API gateway	MVP
Multi-campus per tenant	Campus entity under tenant; per-campus admin role	Phase 2
Feature toggles per plan	Azure App Configuration flags per tenant tier	Phase 2

4. User Roles and Portals

Role	Responsibilities	Data Scope
Super Admin	Platform owner. Manages all colleges, billing, system health, feature flags.	All tenants
College Admin	Manages one college: all campuses, staff, students, fee structure, reports.	Own college
Campus Admin	Manages a specific branch: timetable, local staff, local admissions.	Own campus
Faculty / Teacher	Marks attendance, uploads marks, manages assignments, views timetable.	Own classes
Accountant	Fee management, salary processing, financial reports.	Own college
Student	Views results, timetable, fee status, assignments, clubs, marketplace.	Own profile
Parent	Receives alerts, views ward progress, pays fees, raises queries.	Ward's data
Alumni	Posts jobs, mentors students, participates in reunions, sells in marketplace.	Alumni portal
Club Lead	Manages a specific club: events, members, attendance, announcements.	Own club

5. Functional Modules

5.1. Student Management

MVP

Manages the complete student lifecycle from enquiry and admission through graduation, document generation, and alumni transition.

Feature	Description
Admission Workflow	Multi-step form: enquiry, document upload, fee payment, enrolment confirmation.
Student Profile	Personal details, academic history, photo, emergency contact, documents.
ID Card Generation	Auto-generated with barcode/QR, college branding per tenant.
Promotion Engine	Bulk promote students to next semester/year based on result criteria.
Transfer Certificate	Auto-generated TC with digital signature of principal.
Leaving Certificate	Generated on student exit with attendance and conduct record.
Duplicate Documents	Re-issue ID, TC, LC on request with audit trail.
Student Search	Filter by batch, branch, status, course, campus.

Actors: College Admin, Campus Admin, Accountant

5.2. Attendance Management

MVP

Tracks student and staff attendance through multiple input channels with real-time parent alerts on every absence.

Feature	Description
Faculty Web/App Marking	Class-wise attendance marked period by period.
Biometric Integration	ZKTeco / Suprema device SDK for fingerprint and face ID.
QR/Barcode Scan	Student ID card scan via mobile camera or USB scanner.
Absence Alerts	Instant WhatsApp + SMS to parent on first absent mark.
Shortage Warnings	Auto-flag students below 75% attendance threshold.
Leave Application	Student applies for leave; faculty/admin approves.
Attendance Reports	Daily, monthly, student-wise, class-wise, department-wise.
Staff Attendance	Separate tracking for teaching and non-teaching staff.

Actors: Faculty, Campus Admin, Student, Parent

5.3. Academic Management

MVP

Manages all academic operations including scheduling, syllabus, assignments, internal assessments, and result generation.

Feature	Description
Timetable Builder	Drag-and-drop schedule builder per batch and campus.
Syllabus Upload	Faculty uploads chapter-wise syllabus PDFs per subject.
Assignment Management	Create, distribute, collect, and grade assignments online.
Internal Marks Entry	Faculty enters CIE/mid-term marks with validation rules.
Exam Scheduling	Exam date sheet creation, room allocation, invigilator assignment.
Hall Ticket Generation	Auto-generated hall ticket with exam schedule and rules.
Result Processing	Grade calculation based on configurable grading scheme.
Marksheet / Report Card	Downloadable PDF with college branding and digital seal.
Academic Calendar	Institution-wide calendar: holidays, exam dates, events.

Actors: Faculty, Exam Cell, Students, College Admin

5.4. Fee Management

MVP

Complete fee lifecycle management with Indian payment gateway integration, automated reminders, and granular financial reporting.

Feature	Description
Fee Structure Setup	Configurable: per course, per semester, per category (SC/ST concession).
Fee Challan Generation	PDF challan with QR code for offline bank payment.
Online Payment	Razorpay and PayU integration; UPI, card, net banking.
Late Fine Calculation	Auto-calculated fine after due date with configurable rule.
Payment Receipts	Instant receipt via email, WhatsApp, and in-app.
Fee Defaulters Report	List of students with pending dues, filterable by amount/date.
Scholarship Management	Apply fee waivers and scholarship amounts per student.
Refund Processing	Partial/full refund workflow with approval chain.
Family Fee Report	Consolidated view for parent with all wards on one report.

Actors: Accountant, College Admin, Student, Parent

5.5. HR and Staff Management

Phase 2

Manages the entire staff lifecycle: onboarding, payroll, leaves, and appraisals.

Feature	Description
Staff Profiles	Personal, academic qualification, contract type, department.
Payroll Engine	Salary structure: basic, HRA, DA, PF, PT deductions; payslip PDF.
Leave Management	Leave types (CL, SL, EL), apply-approve workflow, balance tracker.
Staff Attendance	Biometric or app-based; links to payroll deduction on LOP.
Appraisal Records	Annual performance reviews with rating and remarks.
Experience Certificate	Auto-generated on resignation/retirement.

Actors: HR Admin, Principal, Staff Members

5.6. Communication and Alerts

MVP

Multi-channel communication layer ensuring all stakeholders receive timely, relevant notifications.

Feature	Description
Push Notifications	Mobile app push via Azure Notification Hubs.
WhatsApp Alerts	Meta WhatsApp Business API: attendance, fee, results, events.
SMS Gateway	MSG91 / Textlocal for OTP, fee reminders, exam alerts.
Email Notifications	Transactional emails via Azure Communication Services.
Announcement Board	College-wide, campus-wise, or class-specific notice board.
Internal Messaging	Staff-to-staff and staff-to-student secure messaging.
Parent App Inbox	All alerts consolidated in parent's notification inbox.

Actors: All Roles

5.7. Reports and Analytics

Phase 2

Real-time dashboards and exportable reports for data-driven decision making at college and platform level.

Feature	Description
Super Admin Dashboard	Platform-wide: tenant count, revenue, active users, storage.
College Dashboard	Admissions, attendance trend, fee collection rate, exam results.
KPI Widgets	Configurable widgets with drill-down capability.
Attendance Analytics	Heatmaps, defaulter trends, department comparison.
Financial Reports	Income vs expense, category-wise collection, MIS report.
Academic Performance	Subject-wise pass/fail rate, top performers, at-risk students.
Export Options	PDF, Excel, CSV export for all reports.
Scheduled Reports	Auto-email weekly/monthly reports to principal/management.

Actors: Super Admin, College Admin, Principal, Accountant

5.8. Multi-Tenancy and Super Admin Panel

MVP

Central control plane for the platform owner to manage all colleges, subscriptions, and system configuration.

Feature	Description
College Onboarding Wizard	10-minute setup: name, logo, admin credentials, subdomain.
Subscription Management	Plan assignment, upgrade/downgrade, renewal tracking.
Billing and Invoicing	Auto-generated invoices; Razorpay subscription webhook handling.
Tenant Health Monitor	Per-tenant: storage usage, API call volume, last login.
Feature Flag Control	Enable/disable modules per plan tier.
Support Ticket System	College admins raise tickets; platform team resolves.
Bulk Announcements	Platform-wide notices to all college admins.
Audit Logs	All super-admin actions logged with timestamp and IP.

Actors: Super Admin

5.9. Mobile Application

Phase 2

A single cross-platform mobile app serving all user roles with role-based UI rendering. Android-first; iOS in Phase 2.

Feature	Description
Role-Based UI	Login role determines which screens and features are shown.
Offline Attendance Sync	Attendance marked offline; synced when connectivity restored.
Biometric Login	Fingerprint/Face ID authentication on device.
Fee Payment In-App	Razorpay checkout embedded in mobile app.
Push Notifications	Real-time alerts for all relevant events per role.
Digital ID Card	Student ID with QR code viewable offline.
Assignment Submission	Students upload assignment files directly from phone.
Result Viewing	Marksheets and grade cards downloadable as PDF from app.
Platform	React Native (Android + iOS); deployed to Play Store and App Store.

Actors: All Roles

5.10 Clubs and Societies Management

Enables colleges to digitally manage all student clubs, societies, and extracurricular groups. Builds a rich co-curricular profile for every student.

Feature	Description
Club Creation	College Admin creates clubs: Coding, Debate, Sports, NSS, Cultural, Robotics, etc.
Student Enrollment	Students apply via portal/app; club lead approves or rejects with reason.
Club Lead Role	Faculty or senior student designated as club lead with management permissions.
Event Management	Create events, publish schedule, accept RSVPs, send reminders.
Event Attendance	Mark attendance per event; attendance added to student's co-curricular profile.
Announcements	Club-specific notice board; members notified via push/WhatsApp.
Achievements	Log trophies, certificates, wins at inter-college competitions.
Photo/Video Gallery	Event media uploaded to Azure Blob; viewable in app gallery.
Club Leaderboard	Top clubs ranked by member participation rate and event frequency.
Co-Curricular Report	Student receives club activity summary; college admin sees overall participation.
Certificate Generation	Auto-generate participation/achievement certificates with club branding.

Actors: College Admin, Club Lead, Student

Phase: Phase 2

5.11 Alumni Management

Maintains a living alumni network that benefits current students through mentorship, placements, and industry exposure. Alumni of every batch and branch are onboarded and kept engaged through the platform.

Feature	Description
Alumni Self-Registration	Graduates register after passing out; verified against student records by admin.
Batch and Branch Directory	Browse alumni by year of passing, department, campus, and company.
Alumni Profile Pages	Current designation, employer, LinkedIn URL, achievements, skills.
Job Board	Alumni post job/internship openings exclusively for current students.
Mentorship Programme	Students request 1-on-1 mentorship; alumni accept, schedule sessions.
Events and Reunions	Alumni-only events, batch reunions, guest lectures, webinars.
Donation Portal	Optional giving-back portal; alumni donate to college development fund via Razorpay.
Success Story Spotlights	Admin publishes featured alumni achievements on college portal and social media.
Alumni Newsletter	Monthly email and WhatsApp digest: college news, job openings, events.
Placement Statistics	Super admin and college admin view alumni placement data by batch and branch.
Verification Badge	Verified alumni badge on profile after admin cross-check with graduation records.

Actors: College Admin, Alumni, Student, Super Admin

Phase: Phase 3

5.12 Marketplace

A campus commerce layer enabling peer-to-peer trade, official college merchandise, and curated local vendor listings — all within the platform ecosystem.

5.12A Student-to-Student Marketplace

Feature	Description
List Items for Sale	Used textbooks, lab coats, calculators, notes, electronics.
Category Filters	Books, Electronics, Stationery, Lab Equipment, Clothing, Notes/PDFs.
In-App Chat	Buyer contacts seller via Azure Web PubSub real-time chat.
Make Offer	Buyer can negotiate price; seller accepts/declines.
Status Toggle	Seller marks item as Available / Reserved / Sold.
Seller Ratings	Buyer rates seller after transaction; ratings visible on profile.
Report Listing	Flag inappropriate or counterfeit listings for admin review.

5.12B College Official Store

Feature	Description
Merchandise	College hoodies, diaries, lanyards, branded stationery.
Uniform Orders	Standardised uniform purchase per course with size selection.
Study Material Store	Faculty-authored notes and e-books sold as downloadable PDFs.
Integrated Checkout	Razorpay/PayU payment inside app; order confirmation receipt auto-sent.
Order Management	Admin fulfils orders, updates status; student receives delivery notification.
Revenue Report	Monthly store earnings report for college accountant.

5.12C Vendor and Sponsor Listings (Premium)

Feature	Description
Local Business Listings	Nearby hostels, tiffin services, stationers, coaching centres.
Paid Banner Slots	Vendors pay for banner placement on student feed.
Discount Coupons	Verified vendors push exclusive discount codes to students.

Actors: Students, Alumni, College Admin, Accountant, Vendors

Phase: Phase 3

6. Technical Architecture (Azure)

The platform is built entirely on Microsoft Azure. All infrastructure is defined as code using Terraform. The architecture follows a microservices-oriented, API-first design with per-tenant data isolation.

Layer	Component	Azure Service	Notes
Frontend	Web Portal (React/Next.js)	Azure Static Web Apps	Global CDN, auto SSL
API Gateway	API Management	Azure APIM	Rate limiting, tenant routing, JWT validation
Backend	REST Microservices	Azure App Service / AKS	.NET Core or Node.js
Auth	Identity and Access Management	Azure Entra ID B2C	Multi-tenant SSO, social login
Database	Relational Data (per-tenant schema)	Azure SQL / PostgreSQL	Isolated schema per college
File Storage	Documents, Photos, Certificates	Azure Blob Storage	Per-tenant containers; CDN-served
Real-time	Marketplace Chat, Live Alerts	Azure Web PubSub	WebSocket-based real-time messaging
Search	Alumni Directory, Marketplace	Azure Cognitive Search	Full-text search with filters
Notifications	Push Notifications	Azure Notification Hubs	Android + iOS fan-out
Notifications	SMS and WhatsApp	Twilio / MSG91	Indian carrier-grade SMS delivery
Payments	Fee, Store, Donations	Razorpay / PayU	UPI, card, net banking, EMI
Monitoring	Observability, Alerts	Azure Monitor + App Insights	KQL dashboards, anomaly alerts
CI/CD	Deployment Pipelines	Azure DevOps	Blue-green deployments
IaC	Infrastructure as Code	Terraform	All resources version-controlled
Key Management	Secrets, Certificates	Azure Key Vault	Managed identity access only
Backup	Data Protection	Azure Backup + GRS	30-minute RPO; geo-redundant

7. Non-Functional Requirements

Category	Requirement	Target
Availability	Platform uptime SLA	99.9% per month
Performance	API response time (95th percentile)	< 400 ms
Scalability	Concurrent active users per tenant	5,000+
Scalability	Total tenants without re-architecture	1,000+
Security	Data encryption in transit	TLS 1.2 minimum
Security	Data encryption at rest	AES-256
Security	VAPT before each major release	Zero critical findings
Compliance	Indian IT Act 2000 and PDPB compliance	Mandatory
Backup	Recovery Point Objective (RPO)	30 minutes
Backup	Recovery Time Objective (RTO)	2 hours
Usability	Mobile app onboarding (new user to first action)	< 3 minutes
Maintainability	Zero-downtime deployments	Blue-green via Azure DevOps

8. Integration Requirements

System	Provider	Module(s) Used In	Priority
Payment Gateway	Razorpay + PayU	Fee, Marketplace, Alumni Donations	MVP
WhatsApp Business API	Meta (Official BSP)	Alerts, Newsletters, Fee Reminders	MVP
SMS Gateway	MSG91 / Textlocal	OTP, Attendance, Fee, Exam Alerts	MVP
Biometric Device SDK	ZKTeco / Suprema	Attendance (Student and Staff)	Phase 2
Video Conferencing	Zoom / Google Meet API	Alumni Webinars, Online Classes	Phase 3
Accounting Export	Tally ERP	Fee, HR Payroll, Store Revenue	Phase 3
Google Calendar	Google Calendar API	Timetable, Academic Calendar	Phase 3
LinkedIn	LinkedIn API	Alumni Profile Import	Phase 3

9. Business and Pricing Model

Tier-based SaaS pricing targeting all college sizes in India, from small polytechnics to large multi-campus universities.

Plan	Target	Price	Key Limits	Modules
Free	Small colleges	Free forever	Up to 100 students, 1 campus	Core modules only (no mobile app, no marketplace)
Starter	Mid-size college	Rs. 2,000/month	Up to 500 students, 1 campus	All MVP modules + mobile app
Standard	Growing college	Rs. 5,000/month	Up to 1,500 students, 2 campuses	MVP + Phase 2 modules + Clubs
Premium	University / chain	Rs. 10,000/month	Unlimited students, 5 campuses	All 12 modules including Alumni and Marketplace
Branded App	Any plan	Rs. 50,000 one-time	Dedicated app with college name and logo	Published on Play Store + App Store
Enterprise	Large university	Custom quote	Unlimited, dedicated infrastructure	Custom SLA, dedicated support

10. MVP and Phase Roadmap

Phase	Modules	Target Duration	Milestone
MVP	Multi-Tenancy, Student Mgmt, Attendance, Fee Mgmt, Communication, Mobile App (Android)	0 - 4 months	First 10 paying colleges live
Phase 2	Academic Mgmt, HR/Staff, Reports/Analytics, Clubs and Societies, Mobile App (iOS)	5 - 9 months	50 colleges, Rs. 2.5L MRR
Phase 3	Alumni Management, Marketplace, Vendor Listings, Tally Integration, Localisation (Hindi, Telugu)	10 - 15 months	200 colleges, product-market fit

Complete Module Summary

#	Module	Phase	Mobile	API
1	Student Management	MVP	Yes	REST
2	Attendance Management	MVP	Yes	REST + WebSocket
3	Academic Management	MVP	Yes	REST
4	Fee Management	MVP	Yes	REST + Razorpay
5	HR and Staff Management	Phase 2	Yes	REST
6	Communication and Alerts	MVP	Yes	Event-driven
7	Reports and Analytics	Phase 2	Yes	REST + KQL
8	Multi-Tenancy / Admin Panel	MVP	No	REST + Admin
9	Mobile Application	Phase 2	Native	React Native
10	Clubs and Societies	Phase 2	Yes	REST
11	Alumni Management	Phase 3	Yes	REST + Search
12	Marketplace	Phase 3	Yes	REST + WebSocket

11. Glossary

Term	Definition
SaaS	Software as a Service — cloud-hosted software accessed via browser or app.
Multi-Tenancy	A single platform instance serving multiple colleges, each with isolated data.
Tenant	One college (or institution chain) registered on the platform.
APIM	Azure API Management — gateway controlling all backend API access.
AKS	Azure Kubernetes Service — managed container orchestration for microservices.
Entra ID B2C	Microsoft Azure's customer identity platform for external user authentication.
RPO	Recovery Point Objective — maximum acceptable data loss period (30 minutes here).
RTO	Recovery Time Objective — maximum acceptable downtime after failure (2 hours here).
VAPT	Vulnerability Assessment and Penetration Testing — security audit before launch.
KQL	Kusto Query Language — used in Azure Monitor and Application Insights.
IaC	Infrastructure as Code — all Azure resources defined and deployed via Terraform.
MRR	Monthly Recurring Revenue — key SaaS business metric.
BSP	Business Solution Provider — authorised WhatsApp API reseller.
PDPB	Personal Data Protection Bill — India's upcoming data privacy legislation.
GRS	Geo-Redundant Storage — Azure storage replicated across geographic regions.

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